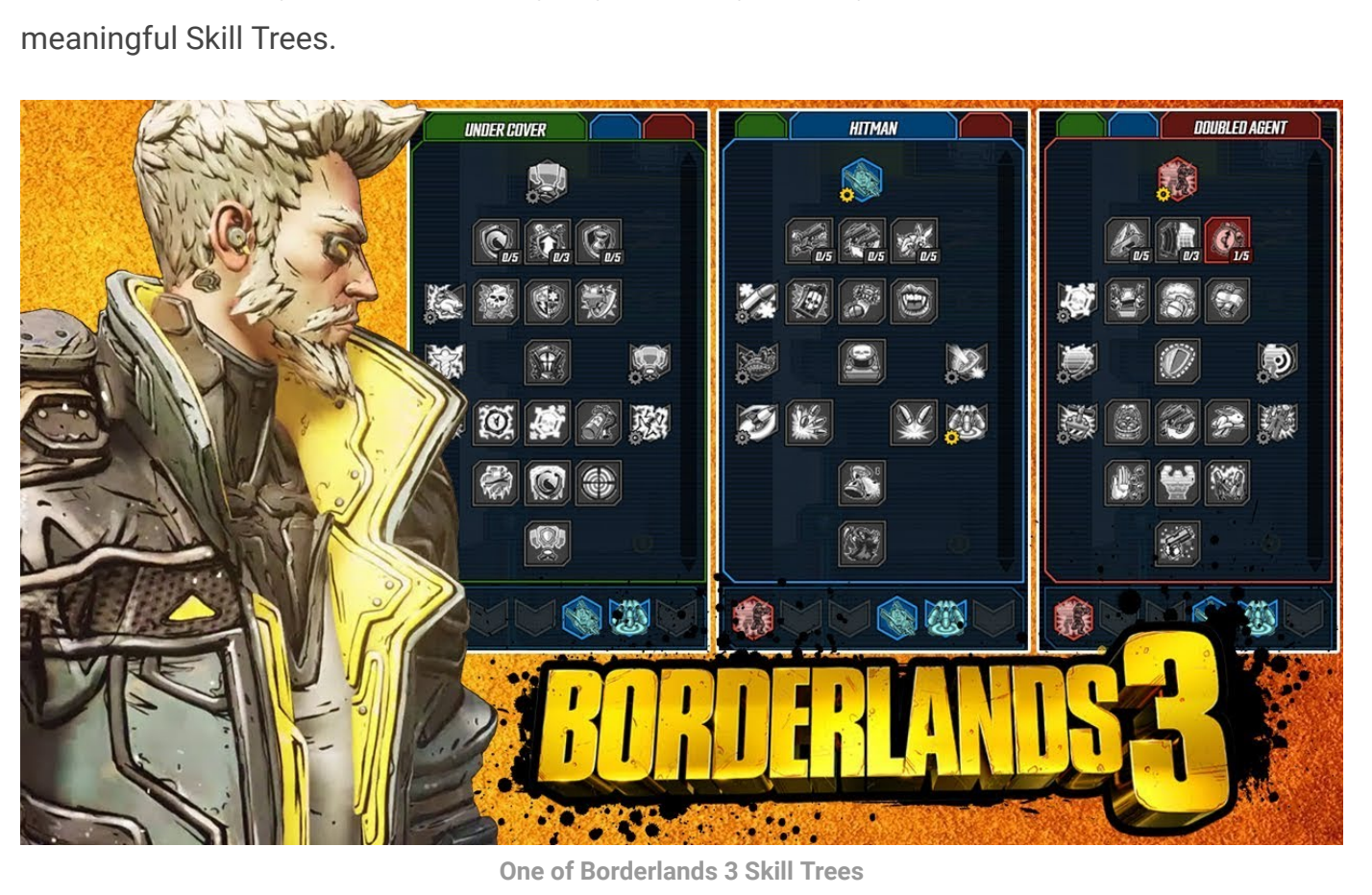
Keys to Meaningful Skill Trees

Nicolas Kraj

Skill trees… One of the **most ubiquitous game systems** nowadays. Some of you love them, most of you believe it's a plague spreading in every single game you play. And indeed, they are found everywhere. In RPGs of course, or strategy games, but also in more peculiar genres: do fast-paced shooters like *Doom*, or single-player action adventures like *Mirror's Edge* and *The Last of Us*, really need a skill tree?

I am convinced you can all find a ton of underwhelming skill trees. Badly crafted, with boring and uninteresting abilities, frustrating gating of what should be common sense skills and given to the players from the get-go… Games that clearly could have been as good without this artificial progression system or worse: **games that would have been better without it!**

This article started as an open discussion on our Patrons Discord and opened enough topics that I decided to lay down some of my Keys to help you study, design, and iterate on meaningful Skill Trees.

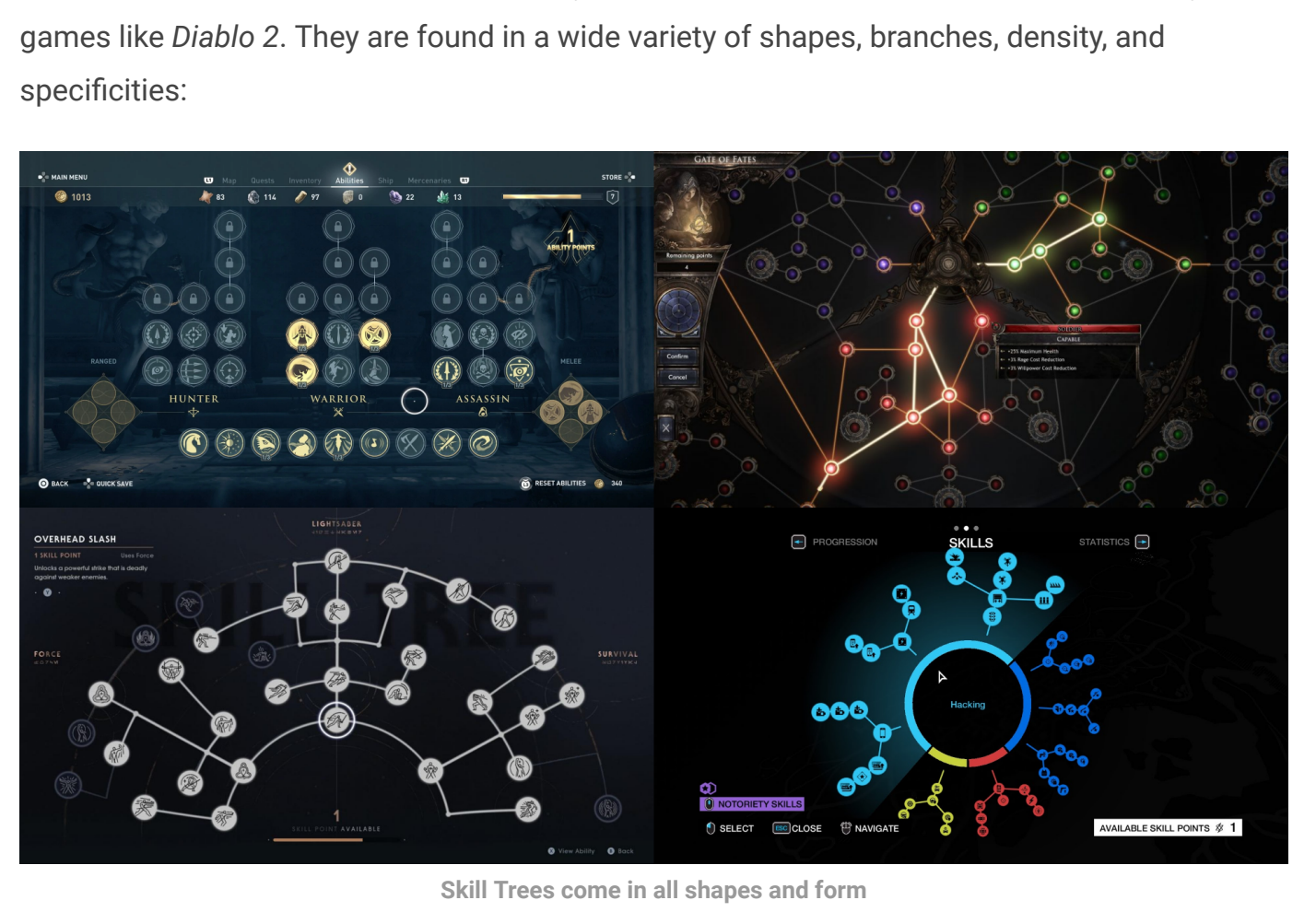


One of Borderlands 3 Skill Trees

This topic is wide and controversial. So **I count on you to bring your own experience, perception, and approach** in the comments or even join our community for some direct discussions!

Skill "Trees"

We will need to define our topic a bit more. First of all, the word "Skill Tree" is pretty limiting: the concept comes from the origins of physical RPG games and has been popularized by games like *Diablo 2*. They are found in a wide variety of shapes, branches, density, and specificities:



Skill Trees come in all shapes and form

A lot of these systems do not look like trees anymore, but this word brings an interesting approach to define our topic:

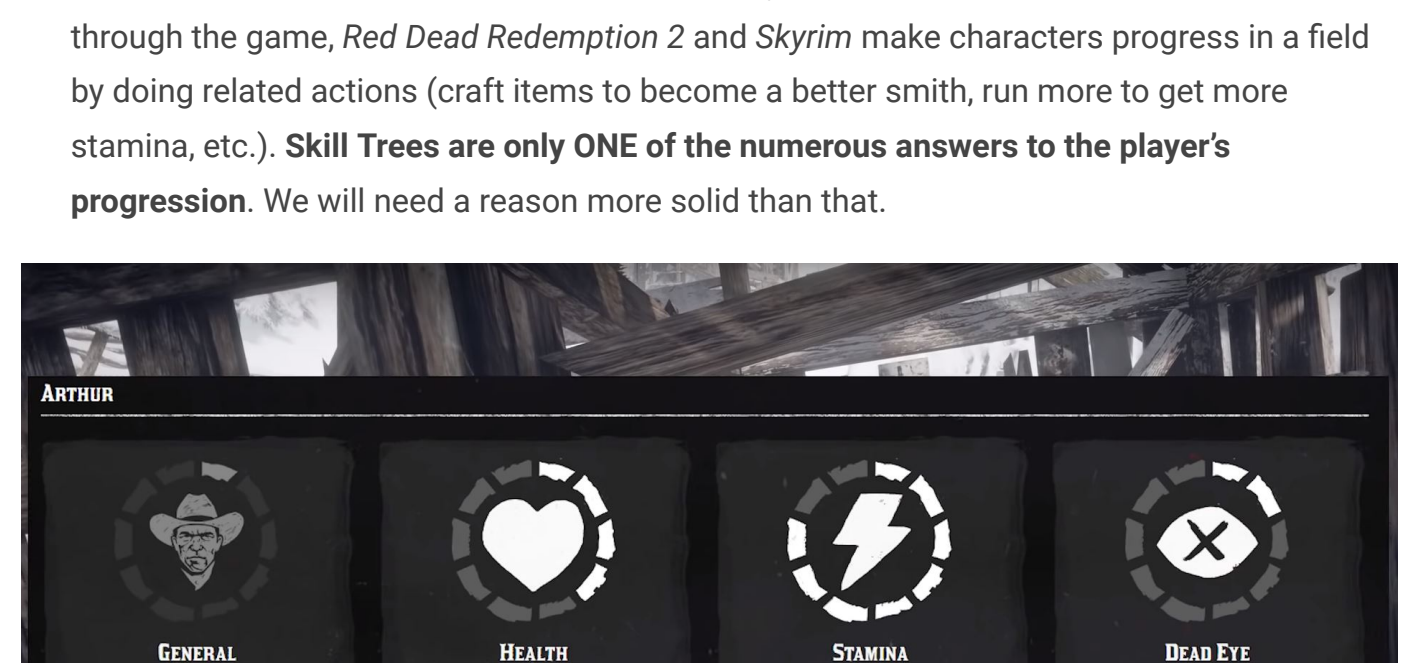
> Skill Trees are progression systems, starting from (a) common basic trunk(s), and expending into more specialized, micro skills.

This somehow broad definition will actually be very helpful when deciding if we should, or not, consider having a Skill Tree in a game.

Does your game need a Skill Tree?

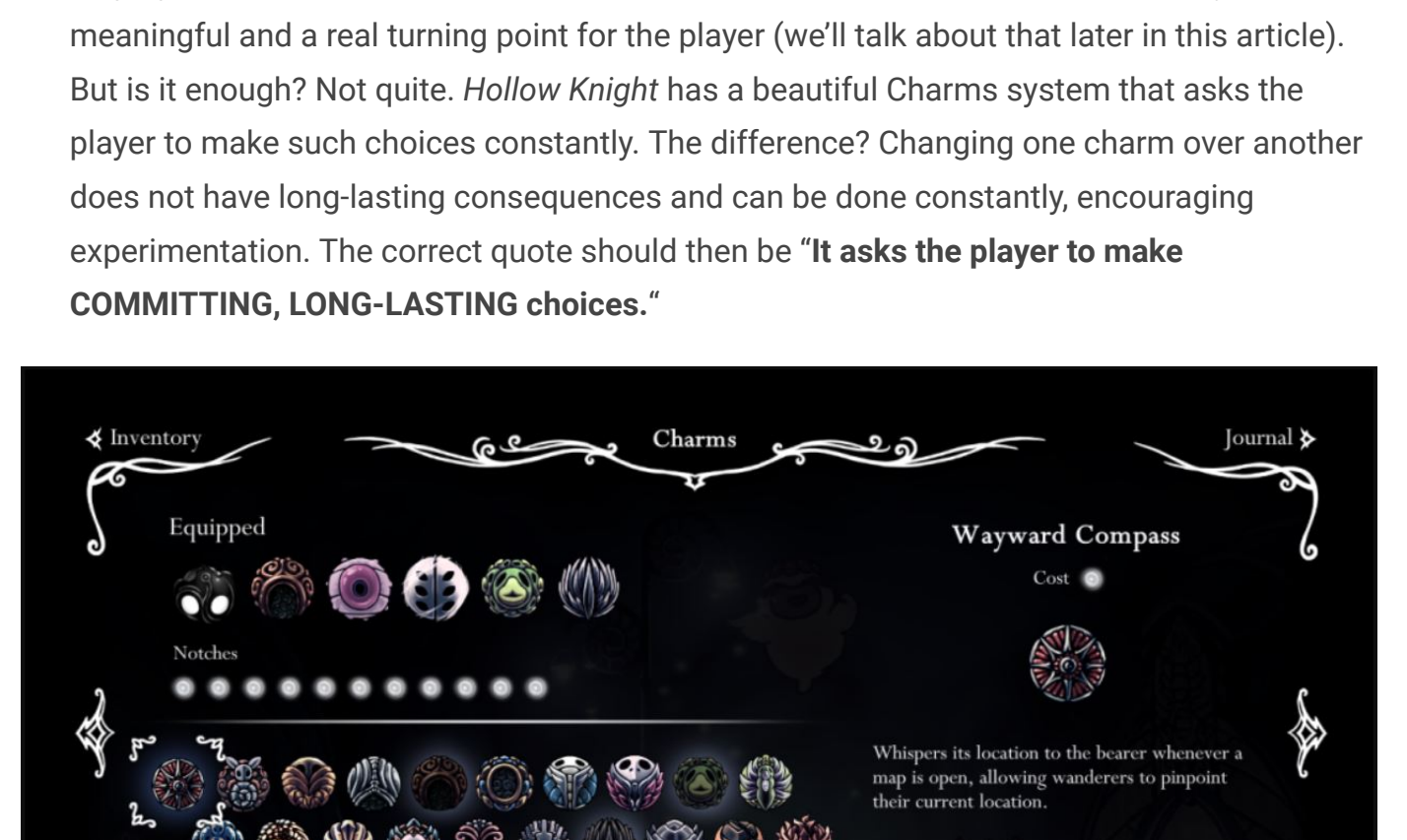
I believe this is the first question any developer (or player studying a game) needs to ask himself. And it is **a question surprisingly difficult to answer**. Let's look at some of the reasons why a game should have one, and see if they present a valid point:

- "**Other games have one**". Kind of a no-brainer, it is **obviously a bad reason**. And yet it is very often included by default on the "must-have" checklist of big productions for this very reason. But we are designers, we can do better than that.
- "**It adds density and playtime**". This one is a bit tougher to answer. At first glance, it seems to be a great reason to have one. But this is not a player-centric reason, and thus a wrong one. **Playtime increase is only a consequence of the system**. A very welcomed one, but not reason enough to force-feed your players with a full-fledged Skill Tree (another quote could have been "**It supports replayability**": the same answer applies).
- "**It gives a tangible sense of progression**". Now we start to talk! Indeed Skill Trees will support a player's solid sense progression. But other progression systems would work very well here too: *Metroid* gives new skills as the player uncovers the world and progresses through the game, *Red Dead Redemption 2* and *Skyrim* make characters progress in a field by doing related actions (craft items to become a better smith, run more to get more stamina, etc.). **Skill Trees are only ONE of the numerous answers to the player's progression**. We will need a reason more solid than that.



Red Dead Redemption

- "**It's a powerful onboarding tool**". Another strong argument, one beautifully illustrated by the latest God of War: the game features a substantial amount of fighting moves, many of them requiring a real mastery of the system to be used and chained. By letting the player get accustomed to a restricted moveset, containing easy-to-use, polyvalent abilities, the developers ensure that the player will learn the game at a correct pace and progress towards the advanced skills confidently. Yet, this is **another valid reason fitting any well-designed progression system**, not only Skill Trees.
- "**It asks the player to make choices**". Yes. This is the core of what makes a Skill Tree a unique progression system. It asks the player to **make constant decisions regarding his playstyle and character development.** Choosing a skill over another should always be meaningful and a real turning point for the player (we'll talk about that later in this article). But is it enough? Not quite. *Hollow Knight* has a beautiful Charms system that asks the player to make such choices constantly. The difference? Changing one charm over another does not have long-lasting consequences and can be done constantly, encouraging experimentation. The correct quote should then be "**It asks the player to make COMMITTING, LONG-LASTING choices.**"


Inventory
Charms
Journal
Equipped
Notches
Wayward Compass
Cost
Whispers its location to the bearer whenever a map is open, allowing wanderers to pinpoint their current location.
Equip

Hollow Knight's Charms page

> **Key #1**: A good Skill Tree forces the player to make impacting, commiting, long-lasting choices for the development of his playstyle.

- "**It allows players to create a unique character**". Yes again. This is much related to the previous point, and logical when one looks at the RPG origins of this system. A good skill tree pushes the player to build a unique character, one that **supports self-expression**, one that will **drive his engagement and appropriation**. The player isn't embodying a hard-written character, he is **projecting himself** through this character.

> **Key #2:** A good Skill Tree empowers your players, encourages self-expression, and support your players' projection in the character they play.

This is a very good start! But there are so many other great reasons to have a Skill Tree: it can be completely exposed from the start, tapping into **players' conditioning and scheduled rewards** (the magic of Skinner boxes and dopamine hit), or instead left with its structure only for **teasing on what might come later in the game**. It can be **encouraging risk-taking** or **driving players' commitment** to a specific build and hard choices along the way. It can **encourage long-term strategies** or deny them… The possibilities are endless, and each game needing a Skill Tree should spend the time and energy to tailor its details to the game's unique experience.

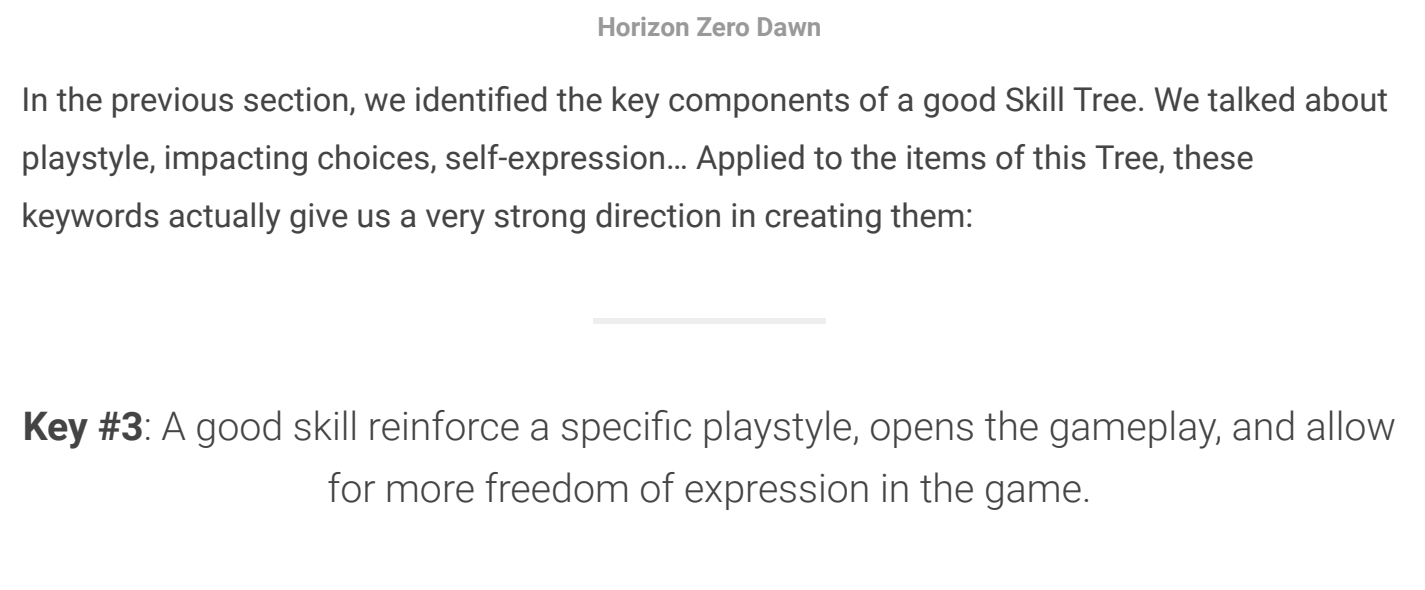
And, of course, we cannot talk about Skill Tree design without talking about the items of said Trees.

Meaningful skills

Designing meaningful skills, in the context of an entire game, is a very wide topic in itself, one that would deserve its own "Keys to" article. That being said, filling a Skill Tree with skills requires some specific attention that plenty of games overlook.

In how many games have you found yourself legitimately bored or uninterested when getting a new skill point to spend? When you feel that filling a skill tree was just a mandatory chore? "+5% XP point on kill", "10% additional fire damage", "+20HP"… To me, this is where the plague lies: **pages and pages of dull skills filling Skill Trees that shouldn't exist**, making your character simply keep up with the artificially increasing difficulty of the game.

But then, what makes a skill interesting? Should using number increases be avoided at all costs? Well, the answer is, as often, not entirely black or white.



Horizon Zero Dawn

In the previous section, we identified the key components of a good Skill Tree. We talked about playstyle, impacting choices, self-expression… Applied to the items of this Tree, these keywords actually give us a very strong direction in creating them:

> **Key #3**: A good skill reinforce a specific playstyle, opens the gameplay, and allow for more freedom of expression in the game.

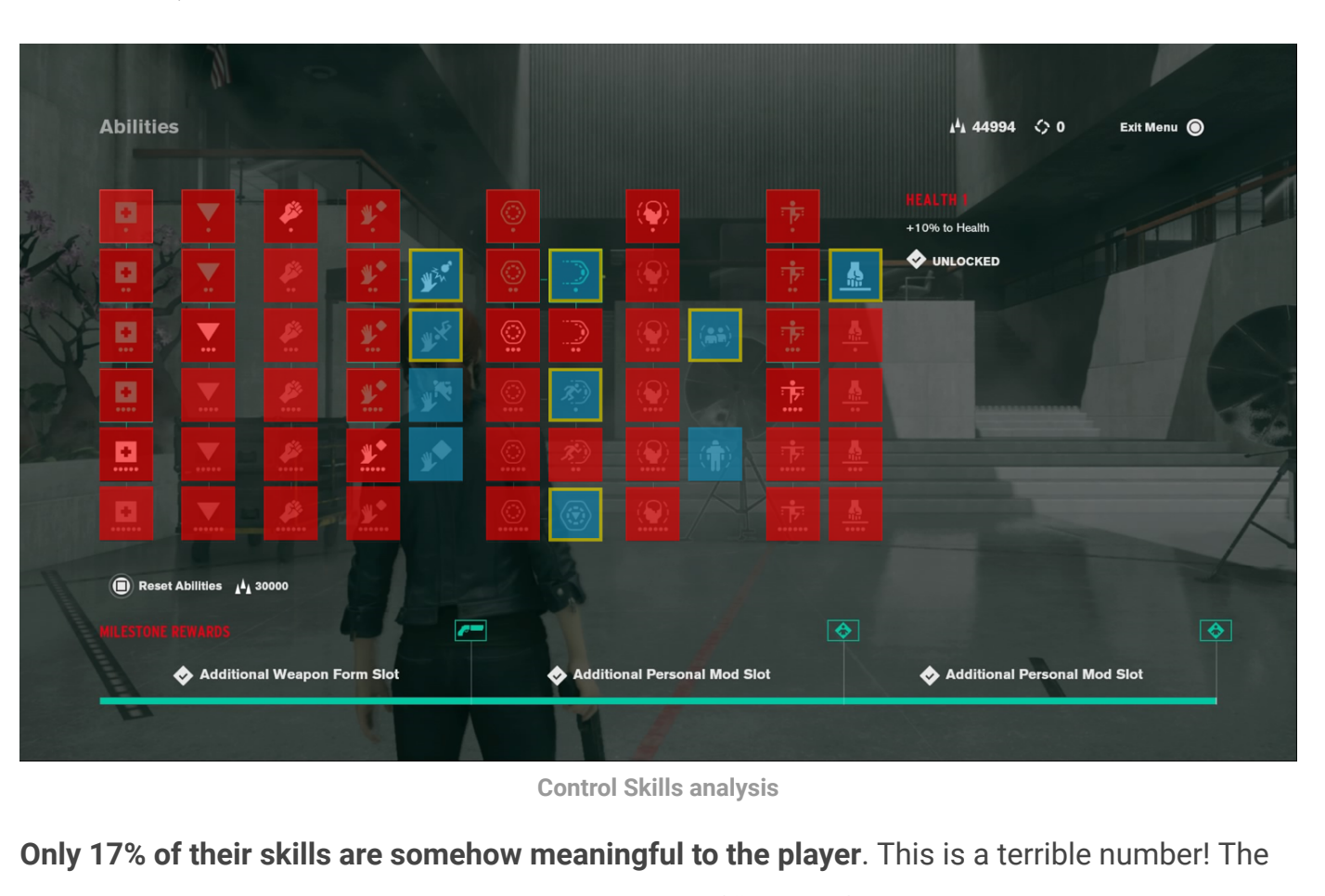
Let's try applying this rule over a few examples and see how it holds up:

- "**+5% XP points on kill**": the player was already killing enemies. He was also already earning XP points on kills. This skill does not bring anything new. It's not a meaningful skill.
- "**You can now assassinate Elite enemies**": The assassination mechanic obviously already existed but with limitations. This skill opens new possibilities for this specific skill that may create new opportunities for the player. It's an OK skill.
- "**Revive dead enemies to fight alongside you**": now this is different! It opens new possibilities for the player, and new risk-reward situations in a fight, it sells a clear Necromancer fantasy. It's a terrific skill!
- "**+100% enemy stun time**": this one is a lot more difficult to answer. It is based on numbers, yes, but significant ones. Will this increase fundamentally change the player's approach to the stun mechanic? It very much depends on the game.

Here is a little game for you to play next time you find yourself bored to death or terribly excited about a skill tree or getting skill points in a game: check how many of these skills fall into these 3 categories:

- **Bad skill**: does not change the gameplay, reinforce a playstyle, or create new player's behavior or strategies.
- **OK skill**: taps into exciting mechanics, often simply lifting restrictions, allowing players to do a bit more with the tools they already have.
- **Good skills**: reinvents the gameplay, adds tools, changes the game, and opens more ways for the player to express himself.

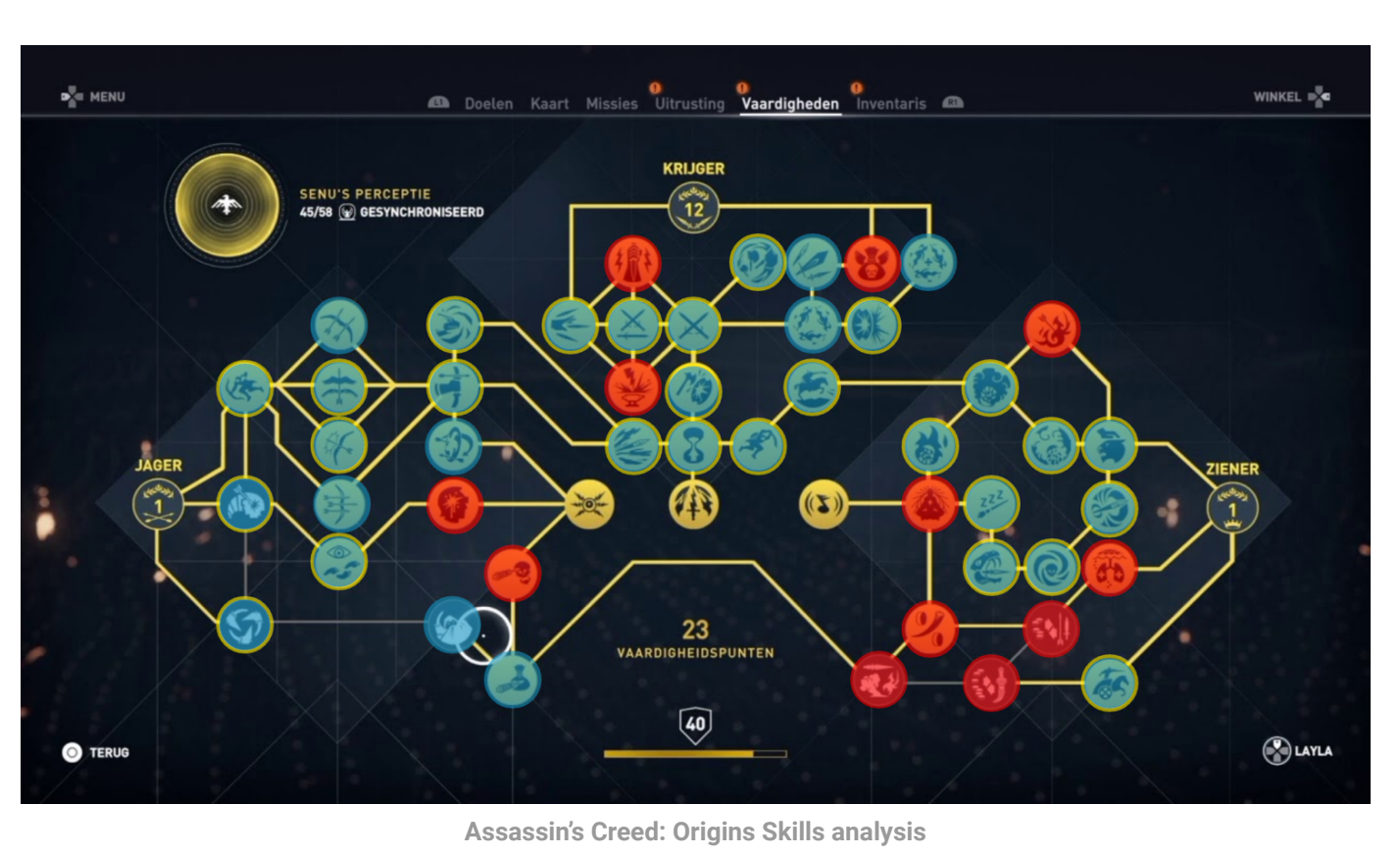
I was playing *Control* recently. Brilliant game (that dark Men-in-Black vibe really got me) but, in my opinion, **the entire progression system could have been cut** and the game wouldn't have been less good. This is their skill tree (Bad skills in red, OK skills in blue, Good skills in blue with gold outline):


Abilities
44994
0
Exit Menu
HEALTH
+10% to Health
UNLOCKED
Reset Abilities
30000
Additional Weapon Form Slot
Additional Personal Mod Slot
Additional Personal Mod Slot

Control Skills analysis

Only 17% of their skills are somehow meaningful to the player. This is a terrible number! The only choices I make along the way are in the order of "+10% life? or 10% energy?" And I usually pick what seems to be what I miss at that specific moment. No excitement, no game-changer, no fresh abilities.

On the other hand, let's have a look at *Assassin's Creed: Origins* (I removed the first 3 given skills and the last 3 that exists for some infinite progression purposes):



Assassin's Creed: Origins Skills analysis

Now we talk about **76% of meaningful skills**. This number will engage players, and embed this system into the core of the game's experience. This is a great Skill Tree from that point of view!

Should you want to try this exercise on the best of the genre, you should check games like *God of War (2018)*, *Dishonored 2*, or, of course, most of the top RPGs. We are getting very close to 100% of meaningful skills. Fun fact, Diablo 2 not only started the trend but also laid down everything we need to know about good Skill Trees:

Diablo 2

Rule of Thumb

I personally have a quick way to assess new skills. It is by no means the only one you should consider, but it helped me more than once and it could help you too!

We are talking about meaningful skills. *Most of the time*, this meaningfulness will express itself through a specific action, and this action will be linked to its verb. So logically, **I try to put a verb in every single one of my skills description**: "cast", "throw", "revive", "heal", "shoot", etc. Bonus point: **If the verb is unique** (i.e. has never been used before for another skill) big chances are that I'm having a killer one under my fingers!

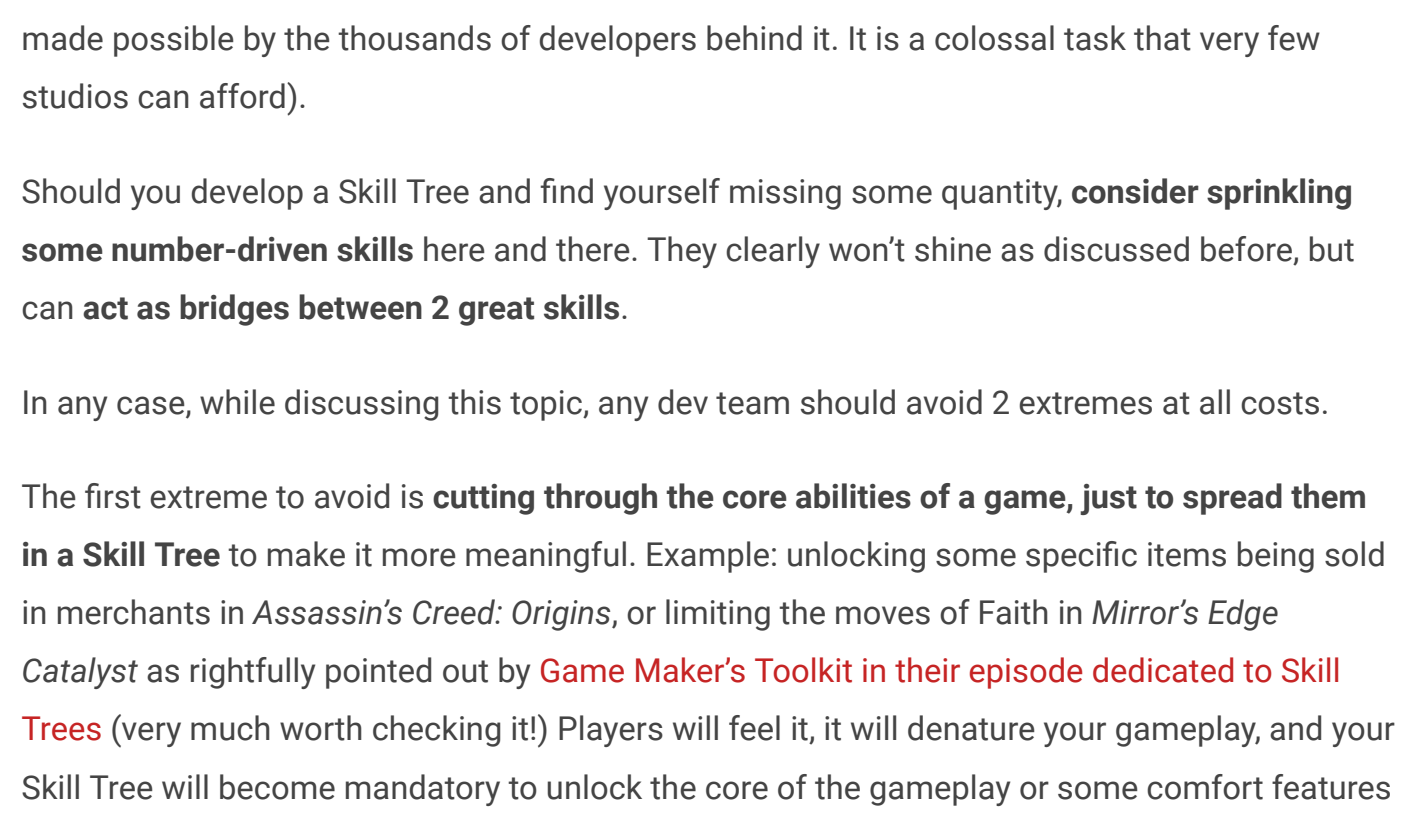
> **Rule of Thumb:** A good skill includes a verb. An awesome skill includes a unique verb.

If you are brainstorming new skills, this approach could open quite a lot of interesting possibilities: **pick interesting verbs, and try to build skills around it**!

Example: "Revert", "Exhaust", "Chain", "Ignore", Prevent", Convert", "Multiply", "Create", "Expand", "Stack", "Change", "Cast", "Buff", "Possess" etc.

Skill Number and extremes

Now a big question is the question of numbers. How many skills are enough? The answer, obviously, is *Path of Exile*:



Path of Exile insane skill tree

Joke aside, it is very difficult to answer this question. There is a **direct correlation between meaningful skills and expensive ones**. Filling hundreds of skills with number increases only takes some UI work, and a double-check on balancing. A new defining ability will add animation sets, gameplay rework, and potentially new systems… So naturally, the greater your skills, the more difficult it will be to produce them (the Skill Tree of *God of War (2018)* is only made possible by the thousands of developers behind it. It is a colossal task that very few studios can afford).

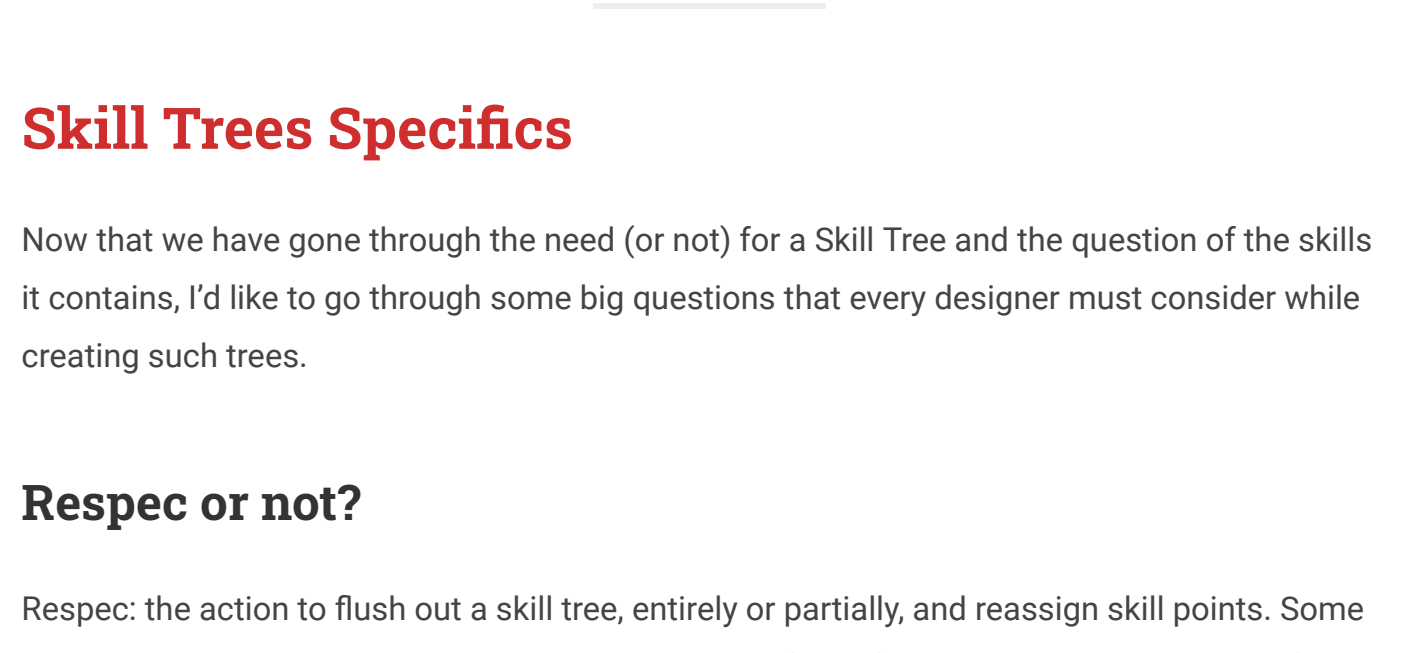
Should you develop a Skill Tree and find yourself missing some quantity, **consider sprinkling some number-driven skills** here and there. They clearly won't shine as discussed before, but can **act as bridges between 2 great skills**.

In any case, while discussing this topic, any dev team should avoid 2 extremes at all costs.

The first extreme to avoid is **cutting through the core abilities of a game, just to spread them in a Skill Tree** to make it more meaningful. Example: unlocking some specific items being sold in merchants in *Assassin's Creed: Origins*, or limiting the moves of Faith in *Mirror's Edge Catalyst* as rightfully pointed out by Game Maker's Toolkit in their episode dedicated to Skill Trees (very much worth checking it!) Players will feel it, it will denature your gameplay, and your Skill Tree will become mandatory to unlock the core of the gameplay or some comfort features that should be given automatically to the players.

The second extreme is somehow difficult to reach: **creating so many skills that they lose their purpose and won't ever be used by the players** (or even be forgotten by the devs themselves). It seems kind of a stretch and yet this is exactly what happened during the development of *Far Cry 3*. There are a lot of assassination-specific skills in this game: from classic takedowns to chain-assassinations, from a ledge, while jumping, etc. etc.

One of them was "**Dual death from below**": a double assassination of 2 enemies standing close to each other at the edge of a ledge, performed with the player coming from below them. As awesome (and expensive to develop) as this skill was, the conditions required were so specific that **the game was released without a single situation to perform it!** Sure, you could always lure enemies and try your best to put them in this specific scenario, but the game legitimately shipped without any LD setup for the player to use this skill (a patch later down the road added a few).


THE HERON
Long range takedowns & mobility
SKILL POINTS AVAILABLE: 4
3995/4000 XP
LB
RB
DUAL DEATH FROM BELOW
REQUIRES: DEATH FROM BELOW TAKEDOWNS (0/1)
Click RS to kill two enemies above on a climbable ledge.
x7.5 XP per kill.

Far Cry 3 most obscure skill

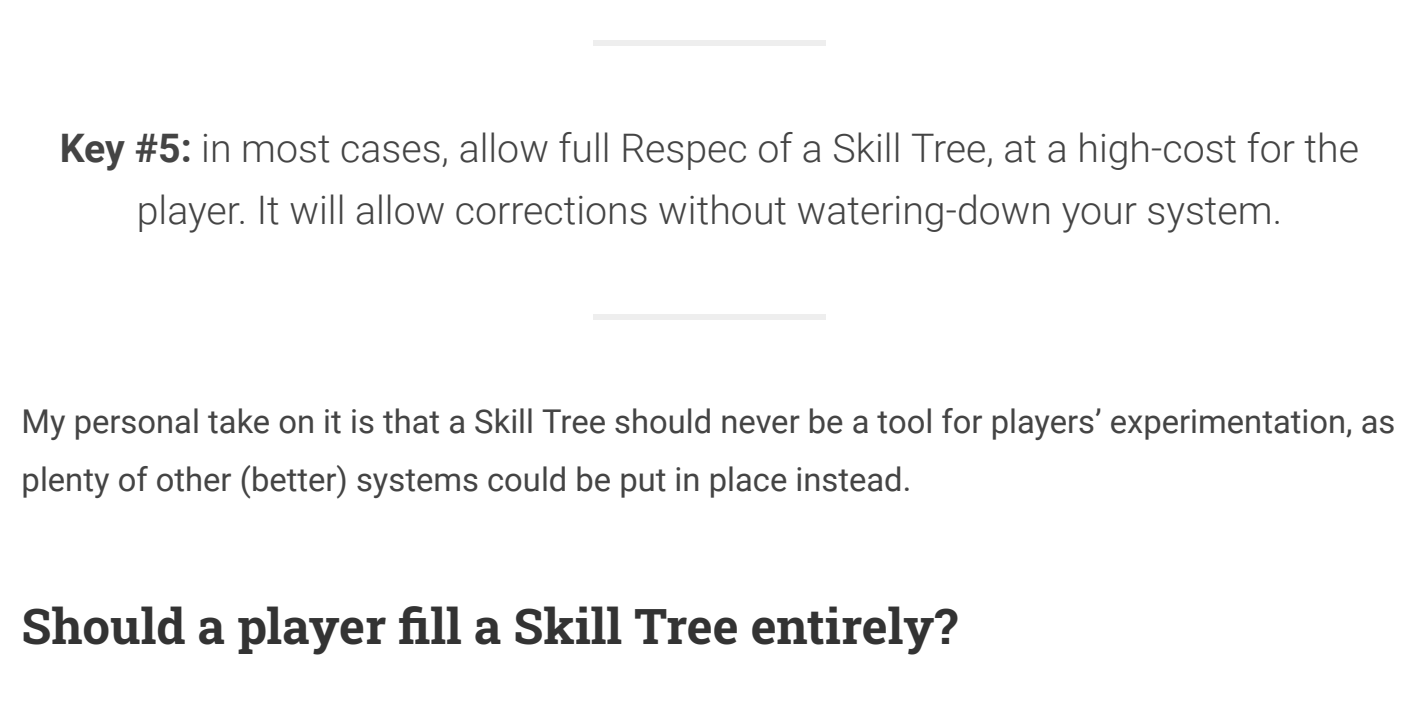
> **Key #4**: A balanced Skill Tree size avoids choice paralysis while still opening clear specialization paths for the player

Skill Trees Specifics

Now that we have gone through the need (or not) for a Skill Tree and the question of the skills it contains, I'd like to go through some big questions that every designer must consider while creating such trees.

Respec or not?

Respec: the action to flush out a skill tree, entirely or partially, and reassign skill points. Some games, like *Borderlands 3* allow cheap, accessible, infinite, full respec. Some, like *Path of Exile*, will only grant players a few tokens to change a single mistake (and considering their skill tree, it's pretty insane). There are 2 big schools of thought here: one allows for experimentation, re-dos, and makes its gameplay shine, taking the risk to undermine the engagement a player will have with his character. The second approach enforces commitment, with the obvious risk of frustrating players who took an unwanted development path: remember that choosing a skill is a bet for the rest of the game: if future challenges aren't clear enough, or the skill can't be tested prior to the investment, players will be prone to make mistakes.



Borderland's Respec cost money, increasing with the Player level

For the vast majority of games, the situation will be the following: you want to encourage role-play, player connection to its character, unicity, and commitment. But you also want to make sure to not have your players rage-quit and have some kind of safeguard in the system.

> **Key #5:** in most cases, allow full Respec of a Skill Tree, at a high-cost for the player. It will allow corrections without watering-down your system.

My personal take on it is that a Skill Tree should never be a tool for players' experimentation, as plenty of other (better) systems could be put in place instead.

Should a player fill a Skill Tree entirely?

Once again, we have **2 schools of thought**: on one side, you have the designers fighting for hard choices, unicity, and specialization. They create trees that can only be filled partially in an

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Design is simply **User-Centered Common Sense**.

entire playthrough. The other side fights for the "100%". The platinum trophy, the completionists (the "we developed those skills, so every single player must see it!").

Again, both sides have strong arguments and each game could approach it differently: RPGs have no reason to allow filling a Tree for example.

Generally, most games nowadays could get away with an in-between:

> **Key #6**: In a non-RPG game, specialize the player for the game duration, then open your systems to allow him to finish his Skill Tree after.

Shadow of the Tomb Raider allows players to fill the Tree entirely

How many ability points per skill?

This question is too dependent on the game itself. Is there a tier system in the skills logic? Are some skills more expensive to force some micro-decisions along the way? Are there some filler-skills? How are the ability points awarded? How often? **Experimentation and iteration** are key here.

Should locked skills be invisible?

Here we start to tap into the unexplored possibility of a Skill Tree: what about **invisible / greyed-out skills when locked**? Sure you'd lose the strategy, the planning, the scheduled-reward dopamine hit, the careful construction of the player's character… But you'd also gain surprises, risk-taking, assumptions… It could be a pretty fresh approach.

But then you'd argue that players are spoiled and would dig wikis, YouTube, and guides… Then make it unique in each playthrough! **Full randomized trees**!

Wait… you are opening too many topics…

I know! And for a good reason: **the industry is throwing Skill Trees in every game and yet hasn't even started tapping into the potential they offer**! GMTK suggests having **skills with negative sides balancing the good ones**. I'd push that further: how about a Skill Tree of negative skills? You start the game fully loaded and, at every level-up, you are forced to choose what will f*** you up: a "Choose your Poison" Skill Tree.

Or a **double-sided Tree**: each Skill point invested gives players a bonus, but enemies another one! It's not only about getting stronger anymore but also preventing enemies from becoming too buffed! I'd call this one "a Balance of Powers".

StormmersGaming suggests playing with **relative numbers and bonuses instead of flat ones**, letting the system loose a bit and creating some insane snowballing effects opportunities (another very good article to read on the topic by the way).

What about **obscure Skills icons and descriptions** that would require players to try and decipher them, running the risk of shooting themselves in the foot if they misunderstand one?

What about **interlinked Skills**? Or some **meta-system** where each skill acquired would be part of rune drawing over the entire Tree (FMA-style game)? Could we restrict tiers of skills differently? Or gain ability points differently? What if we let players swap between talent trees? Have you ever noticed how much the approach of talent trees changed in WoW over the years? This is a crazy interesting example to study!

My point is that **possibilities are endless**, and there are so many paths we haven't explored yet, it would be a shame to just copy what everybody else is doing!

> **Key #7**: Explore uncharted territories, adventure into weird progression systems… So much is still to be found in this domain!

How about you? Do you have wild ideas to add to this list? A different approach to Skill Tree design? Examples of amazing Trees? Let us know in the comments or on our Discord!

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15+ years veteran game designer.
Creator of GDKeys.
Founder of Snackbar Studio and Wonder Goose Studio.

April 5, 2020
Keys
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ADD YOURS

Upper_Krust
February 10, 2021 at 11:06 pm

On the subject of Invisible Skill Trees, I've always liked the tease of having the next potentially unlocked skills visible, but all the ones after that are invisible.

Ena (@InfernalBard)
September 17, 2021 at 6:52 am

Fantastic article, thank you very much!
The one skilltree I'd throw up there as an unexpected positive example are some of the Sims4 expansions. As a vampire , you get to choose both positive "perks" when you reach a new level, but are required to take negative ones as well. And often it's the negative ones that make playing much more interesting, because they tend go much deeper into the core game of time and resource management than the positive skills.
The Get Famous has a similar approach, but the negative traits are assigned based on your Sim's actions.
The one for becoming a magician is probably the most boring skill tree of all of them.

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